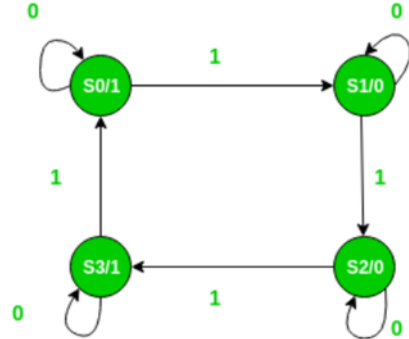


Assignment-3

Q. No.	Question	CO	Bloom
1	Apply the top-down design approach to outline the design steps for a digital stopwatch, from specification to implementation.	CO2	Apply
2	Apply the concept of pipelining to convert a given non-pipelined 4-stage combinational circuit into a pipelined design and identify the additional registers required.	CO3	Apply
3	Apply resource sharing techniques to reduce the number of adders required in a given data path with two sequential addition operations.	CO3	Apply
4	Evaluate a complete case study of a pipelined and resource-shared ALU design, critically analyzing the balance between speed, area, and power.	CO3	Evaluate
5	A Moore FSM has four states (S_0 , S_1 , S_2 , and S_3) with outputs 1, 1, 0, and 0, respectively. Starting from S_0 , determine the sequence of states and outputs for the input sequence: 1, 0, 1, 1, 0.	CO2	Analyze
6	Construct the state transition table from the given Moore machine and analyze the output associated with each state. Assume S_0 as an initial state. 	CO2	Analyze
7	Design a Moore Finite State Machine to detect the binary sequence "1011" in a serial input stream. The FSM should produce an output '1' only when the complete sequence is detected; otherwise, the output should be '0'. Overlapping sequences should also be detected. Draw the state diagram, construct the state table, assign binary state codes, and explain the operation of the FSM.	CO2	Analyze
8	Design a Mealy machine to detect sequence 1101 (overlapping allowed).	CO3	Apply
9	Analyze a pipelined datapath design and calculate the throughput improvement compared to a non-pipelined implementation for a 5-stage pipeline performing 20 instructions. Consider time taken by each stage is 10ns.	CO3	Analyze
10	Design a Moore machine to detect sequence 1001 (overlapping allowed).	CO3	Apply